

field helping aiding doctors in detecting and curing various diseases through image processing.

It would certainly be great if you could learn to develop your Image Recognition system. It is quite possible and that too only little coding is required. It has all been possible due to the evolution of powerful and user-friendly modern day machine learning libraries like Keras. In this article, we are going to train a Convolutional Neural Network using Keras library in Python. Although it is a small Image dataset yet, we will develop a good classifier, which will have an accuracy of about 81%. We are going to use a powerful technique called transfer learning, which helps us increase our accuracy by up to about 95%.

Problem Statement:

First, we will develop a simple image recognition tool using Keras in Python that classifies whether the input image is of a dog or a cat. We will use Convolutional Neural Networks and the technique of Transfer Learning to increase the testing accuracy. Then we will apply the same concepts on a bigger scale to perform some bigger image recognition tasks.

Tools and technologies required:

There are two options to develop the model – either you can develop it on your laptop or train and test it on a server that allows you to use their processors for free such as Kaggle Kernel, or Google Colaboratory (CoLab) kernel. It is recommended to use your personal computer only if it has a Graphical Processing Unit or GPU because training data is a computationally expensive task and requires loads of processing power. The basic tools required for both options are the same. You are required to work with the following libraries in Python:

1. **Numpy:** It is one of the most popular mathematical libraries in Python. It makes computing large, multi-dimensional arrays and matrices very easy and fast. It has a very large collection of high-level mathematical functions that enable you to perform any of the tasks required easily.
2. **Matplotlib:** It is a plotting library for Python. It can be used for plotting a variety of graphs such as line chart, bar chart, histograms and even displaying Images.
3. **scikit – learn:** It is an open source, commercially available Python library that contains efficient tools for data mining and data analysis. We will be using a `train_test_split` function from the `sklearn` library to divide out